NLP Project: Topic Modeling of the 1981 French Legislative Elections

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Abstract

This report details the implementation of NLP techniques to calculate theme distributions across political parties using the Archelec corpus from the 1981 French legislative elections. We describe the data retrieval process, the statistical properties of the corpus, and leverage the BERTopic framework to visualize the intersection of political discourse and party affiliation. Our findings highlight the thematic distinctness of major political entities during this period.

1 Problem Definition

Topic modeling is a technique in NLP that helps to identify themes within large texts. It is primarily used to automatically identify topics in a collection of documents without the need for manual annotation. The goal of this project is to automatically extract the main political themes of the 1981 election and quantify how these topics are distributed across specific political parties. The dataset used is the *Archelec* archive, specifically focusing on the 1981 Legislative election profession de foi (manifestos).

2 Data Retrieval

2.1 Preprocessing

The raw text underwent standard NLP cleaning: removal of punctuation, lowercasing, and the elimination of French stop words.

2.2 Statistical Description

The following metrics describe the processed corpus. There is a total of 3,182 manifestos. The documents average 4138 characters per text before cleaning and 2573 characters after cleaning, which corresponds to 801 tokens before cleaning and 303 after. Regarding the class balance of the parties, the distribution is highly skewed and dominated by major parties like the PS, RPR, and UDF, while dozens of fringe parties contribute almost single-document entries.

3 State of the Art

Topic modeling has evolved from purely statistical counts to deep semantic understanding. Classical methods include Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF). These rely on bag-of-words approaches and often struggle with short texts or nuanced context. Modern approaches rely on neural models such as Top2Vec and BERTopic, which utilize transformer-based embeddings to capture semantic meaning rather than just word frequency.

4 Proposed Model

4.1 Model Selection

For this task, we selected BERTopic, a transformer-based neural topic modeling framework.

4.2 Technical Workflow

BERTopic extracts latent themes from the text through a series of steps. First, documents are converted into dense, high-dimensional semantic vectors. We utilized `dangvantuan/sentence-camembert-base`, a model specifically optimized for the French language, to capture the historical and political nuances of the 1981 electoral texts. Because clustering in high dimensions is computationally expensive and prone to the curse of dimensionality, UMAP (Uniform Manifold Approximation and Projection) is applied. This reduces the vectors to a lower-dimensional space while preserving their local and global geometric structures. Next, HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) groups the reduced vectors into dense clusters, forming the topics. Finally, BERTopic extracts the most representative words for each cluster using a class-based variation of TF-IDF.

4.3 Justification

The choice of BERTopic over classical models like Latent Dirichlet Allocation (LDA) or Non-negative Matrix Factorization (NMF) was driven by specific requirements of our corpus and analytical goals. The first requirement is semantic superiority. Traditional models rely strictly on word frequencies and struggle with synonyms or varied phrasing. By using contextual embeddings like CamemBERT, BERTopic understands the actual meaning of the sentences. It can recognize that two manifestos discussing an economic policy belong to the same theme, even if they use entirely different vocabulary.

The second requirement is class-based analysis for political parties. The primary goal of this project is to analyze theme distributions across different parties. BERTopic's c-TF-IDF is specifically designed for this functionality. It allows us to seamlessly group documents by their political party and extract the unique vocabulary that distinguishes one party's discourse from another.

5 Experiments

The project was implemented in Python. Data manipulation and text extraction were handled using `pandas` and `pathlib`. The core NLP pipeline relied on `Sentence-Transformers` for embeddings, `scikit-learn` (`CountVectorizer`) for text vectorization, and the `BERTopic` library. Visualizations were produced using both Plotly for the top words and `matplotlib` for the thematic distributions.

To capture the specific semantic and syntactic context of the French language, documents were embedded using `dangvantuan/sentence-camembert-base`. This CamemBERT-based model is highly optimized for French sentence-level representations.

Regarding the hyperparameters and configuration, a list of French stop words (`stop_word_fr.txt`) was passed to the `CountVectorizer` to clean the corpus of non-semantic noise prior to clustering. In the BERTopic settings, the language parameter was set to multilingual, and document-topic probability calculations were set to False to optimize memory consumption and execution speed. Finally, to ensure robust statistical representation and readable visualizations, the final analysis was restricted to the 10 most frequently occurring topics (Topics 0 to 9) and the top 10 largest political parties.

Analysis and Conclusion

5.1 Results

Below are the visualizations representing the topic distributions and the thematic clusters identified by the model.

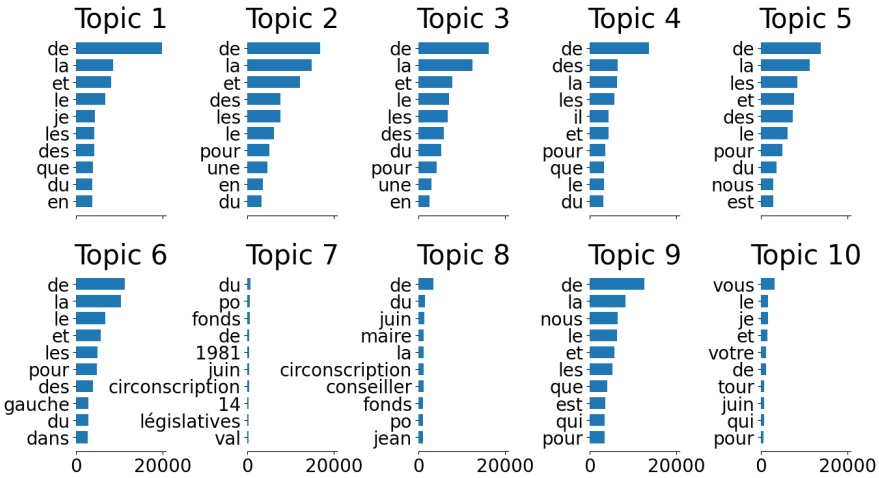

Figure 1: Top words (LDA) without preprocessing

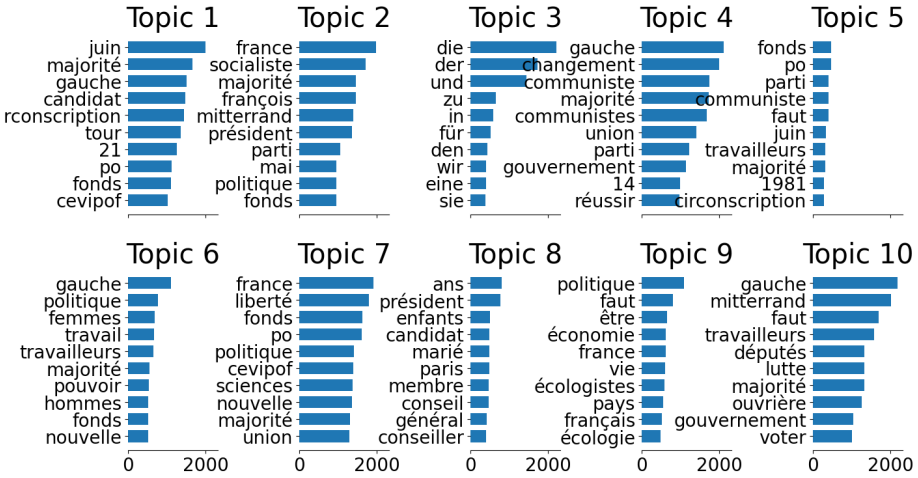

Figure 2: Top words (LDA) with stop words

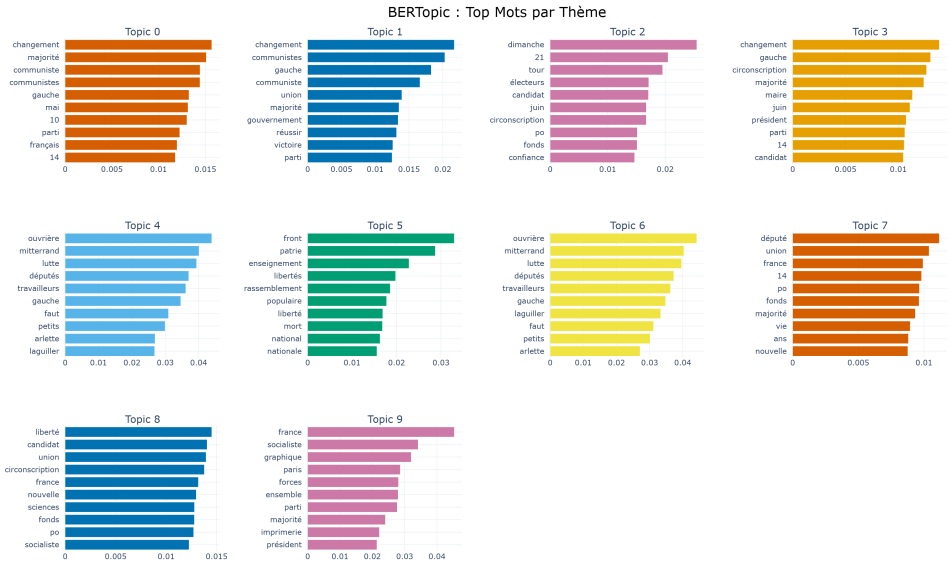

Figure 3: 10 Topics (Bertopic)

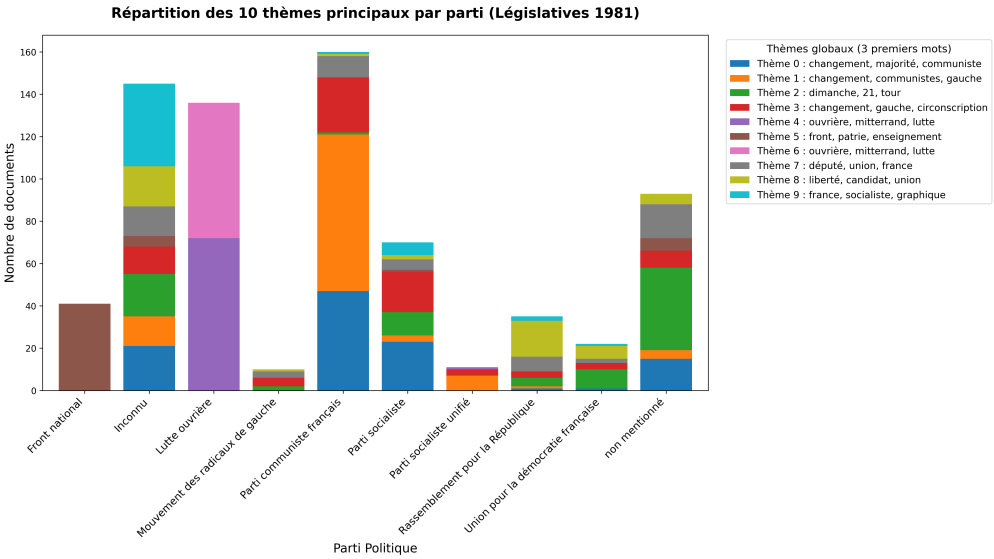

Figure 4: Distribution of Topics per Party: 10 most prominent political groups.

5.2 Analysis

A comparison of the initial baseline models highlights the necessity of the technical workflow. Figure 1 demonstrates that without preprocessing, the generated topics are entirely dominated by non-semantic grammatical noise, with the most frequent words being "de", "la", "et", "le", and "des" across all clusters . While Figure 2 shows that removing stop words yields more relevant political vocabulary (such as "gauche", "majorité", and "candidat") , it still relies on flat frequency counts.

The BERTopic model successfully identified core 1981 themes with much richer semantic grouping, as shown in Figures 3 and 4. The themes are clearly different between parties, and they match what we would expect from them. For example, Lutte ouvriere focuses entirely on topics 4 and 6 (lutte, Mitterand, ouvriere), while Front National focuses on theme 5 (front, patrie, enseignement). Other parties like PS are mixing a lot of different themes. RPR and UDF have the same themes associated to them. It is important to note that often the party is not mentioned (or not known), so themes are diverse. This can be interesting to see to what parties those who do not have a party are the closest to.

5.3 Conclusion

BERTopic proved highly effective for this corpus. Transformer-based embeddings captured the ideological nuances of the 1981 campaign more effectively than traditional frequency-based models. It successfully overcame the limitations of classical bag-of-words approaches like LDA. This semantic superiority allowed the model to understand the actual meaning of the sentences, grouping manifestos by theme even when varied phrasing or different vocabulary was used.

References

[1] Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv preprint arXiv:2203.05794.